CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in integrating LLMs into enterprise solutions, from Generative AI prototypes and agentic workflows to cloud-native deployment on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector built with SAP domain experts, backed by consulting work translating business requirements into AI architectures.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present — Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and translating requirements from SAP domain experts across the German energy sector into the system design.
- Designed the multi-agent orchestration in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% measured against an SAP-system-derived ground truth.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Neo4j, GraphRAG, RAG, FastAPI, MongoDB, Postgres, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 — Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them and translated their business requirements into the AI architecture.
- Designed and deployed an agentic AI application with LangGraph that routes decision-making across vector database queries, general inference and summarization, taking it from prototype to a production-ready, deployed application.
- Built the Python/FastAPI backend and a React frontend, with API testing in Postman, containerization with Docker and deployment to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 — Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval against chemist-provided ground truth, on serverless AWS infrastructure for PrismaChat RAG evaluation (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: LLM Gateway | Python | FastAPI | LLM APIs

- Provider-agnostic API layer over multiple LLM provider APIs with token budgets, streaming, timeouts, retries and fallback between providers, logging latency, token usage and cost for every request.
- Metrics endpoint for cost per request, latency percentiles and evaluation pass rate; tests and evaluation run in CI.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning

- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **Generative AI & LLMs:** LLM integration (RAG, GraphRAG, agents, MCP tools), LangGraph, LangChain, LiteLLM, prompt engineering, QLoRA fine-tuning, evaluation (RAGAS, PytrecEval), Hugging Face Transformers
- **Programming & Testing:** Python, API testing (Postman), SQL, FastAPI, Python microservices, React (beginner)
- **Cloud-Native & DevOps:** Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, CDK), Docker, Kubernetes, Git, GitLab CI/CD
- **Data & ML:** Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas
- **Ways of working:** AI-assisted development with Claude Code, Agile, stakeholder collaboration

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)